COM3021

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

COMPUTER SCIENCE

Examination, May 2023

Data Science at Scale

Module Leader: Dr. Hugo Barbosa

Duration: 2 HOURS

- All questions are compulsory.
- The word estimates are provided as a guidance for the expected length of an effective and concise answer to the questions.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

No electronic calculators of any sort are to be used during the course of this examination.

This is a CLOSED BOOK examination.

Question 1

(a) Describe the difference between a **fault** and a **failure** in the context of systems availability. *(a satisfactory answer will have between 20 and 40 words)*

(5 marks)

(b) Describe three aspects of **hardware faults** that make them potentially **less damaging** to the system availability in comparison with software faults. *(a satisfactory answer will have between 50 and 70 words)*

(10 marks)

(c) When it comes to making decisions related to the design of data-intensive systems, **explain** the trade-off between **availability** and **consistency**, and how choices made based on this trade-off can affect the scalability of said system.

(10 marks)

(a satisfactory answer will have between 70 and 110 words)

(Total 25 marks)

Question 2

- In the context of document models, explain data locality, mentioning **one advantage** and **one disadvantage** of said model in comparison to a traditional relational database. *(a satisfactory answer will have between 10 and 30 words)*

(5 marks)

- In the context of distributed data architectures, explain **two benefits** and **two drawbacks** of combining **replication and partitioning** . *(a satisfactory answer will have between 50 and 80 words)*

(10 marks)

(Total 15 marks)

Question 3

Regarding the programming model MapReduce, describe:

(a) the workflow of a MapReduce job execution.

(5 marks)

(b) **two benefits** and **two limitations** of the MapReduce model.

(8 marks)

(c) Which component of the MapReduce framework is being described in the paragraph below? *(a satisfactory answer will have between 1 and 3 words)*

> *"This function is called once for every input record, and its job is to extract the key and value from the input record. For each input, it may generate any number of key-value pairs."*

(2 marks)

(Total 15 marks)

Question 4

(a) Explain the main difference between synchronous and asynchronous replication in a *leader and followers* architecture. *(a satisfactory answer will have between 20 and 50 words)*

(5 marks)

(b) In a *leader and followers* replication architecture, explain how **read** and **write** operations are served and performed by the replica nodes. *(a satisfactory answer will have between 30 and 60 words)*

(10 marks)

(c) Explain what would be a fully synchronous replication set and what would be a key disadvantage of this configuration. *(a satisfactory answer will have between 40 and 70 words)*

(15 marks)

(Total 30 marks)

Question 5

(a) Reconstruct the matrix using the compressed sparse row (CSR) representation below.

$$
\begin{aligned}
V &= [2, 4, 1, 1, 2, 1, 1, 1, 3, 3, 1, 1] \\
ROW\_INDEX &= [0, 3, 5, 7, 9, 11, 12] \\
COL\_INDEX &= [0, 1, 4, 1, 5, 0, 2, 3, 4, 4, 5, 5]
\end{aligned}
$$

(15 marks)

(Total 15 marks)